WITNESS-1 Preregistration

Quantum-Sourced Authorization Nonces with Verifiable Provenance

Series: WITNESS (first record)
Author: Derek Adam Hone / Remnant Fieldworks Inc.
Preregistration lock date: 2026-07-18 (UTC)
Status: PREREGISTERED — locked before execution

---

1. Claim (single, falsifiable)

A ProofRecord whose challenge nonce is derived from measured QPU output, with the IBM job ID and raw counts embedded, can be:

(a) verified end-to-end against the provider's job record, and
(b) detected as forged when the nonce, job ID, or counts are substituted —

under the tested construction.

---

2. Non-Claims (explicit)

- Does NOT prove quantum randomness superiority over CSPRNG. No Bell test is performed; device bias is present and acknowledged.
- Does NOT prove universal security. Results apply only within the tested construction, circuit model, backend, qubits, calibration snapshot, shot count, and software harness.
- Does NOT constitute a production certification or cryptographic security proof.

---

3. Circuit Specification

- **Qubits:** 8
- **Circuit:** Hadamard on all 8 qubits, measure all — one circuit only.
- **Shots:** 4,000, one job submission.
- **Backend selection:** Least-busy available IBM Quantum backend at submission time. Backend name, calibration snapshot, timestamp, and job ID recorded in results/raw/.
- **QPU time budget:** Estimated seconds. Account must have sufficient remaining time (~2 minutes minimum). If insufficient time remains, STOP — do not purchase time.

---

4. Nonce Construction

```
nonce = SHA-256(
    canonical_json(raw_counts) || job_id || canonical_json(calibration_snapshot)
)
```

- **Canonical JSON:** sorted keys, no whitespace, UTF-8 encoding.
- **Concatenation:** byte-level, no separator between fields.

ProofRecord structure:

```
{
  "nonce":              "<hex string, 64 chars>",
  "job_id":             "<IBM Quantum job ID string>",
  "backend":            "<backend name string>",
  "calibration_hash":   "<SHA-256 hex of canonical_json(calibration_snapshot)>",
  "timestamp_utc":      "<ISO-8601 UTC string>",
  "raw_counts_hash":    "<SHA-256 hex of canonical_json(raw_counts)>",
  "context_id":         "<string — execution context identifier>"
}
```

---

5. Preregistered Test Cases & Pass/Fail Criteria

W1-C1 — Honest Verification

Procedure: Recompute the nonce from stored raw counts + job ID + calibration snapshot. Verify the recomputed nonce matches the ProofRecord nonce exactly. Independently confirm job ID and counts against the IBM provider job record via API.

PASS criterion: Recomputed nonce matches; provider job record confirms job ID and raw counts match stored values. Verification succeeds end-to-end.

Expected outcome: PASS.

W1-C2 — Substitution / Tamper Detection

Procedure: Three independent sub-trials, each altering exactly one field:

- (a) Substitute raw_counts with a different valid-format counts dict; rerun verification.
- (b) Substitute job_id with a different string; rerun verification.
- (c) Substitute calibration_snapshot with a different valid-format dict; rerun verification.

PASS criterion: ALL THREE substitutions are detected as forged (nonce mismatch reported for each). No substitution passes verification.

Expected outcome: PASS (all three tamper attempts detected).

W1-C3 — Cross-Context Replay

Procedure: Present the valid nonce/ProofRecord (from W1-C1) to a second authorization context with a different `context_id`. Use the ARK-457 replay-protection logic (imported as a library module — `src/ark457_replay.py`) to evaluate the replay.

PASS criterion: The replay is DENIED by the replay-protection logic (context_id mismatch detected). The original context_id is accepted; the replay context is rejected.

Expected outcome: DENY (PASS).

---

6. Overall Verdict Rule

PASS iff all three cases meet their stated criteria.

Any single case failing its criterion = overall FAIL, published as-is with full root-cause analysis. A follow-up record (WITNESS-1b) may be registered if warranted.

---

7. Related Works

- ARK Corpus (ExecutionProof series): concept DOI 10.5281/zenodo.21398675 (19 records, 17 experiment IDs, 16 PASS / 2 FAIL / 1 gate-stop)
- ARK-457 (cross-context replay, most directly related): DOI 10.5281/zenodo.21421742 — replay-protection logic used here as library.
- WITNESS-1 is an independent record in a new WITNESS series; it is not a fork of executionproof-testbeds or uip-phase1-testbeds.

---

8. Mandatory Scope Disclaimer

> “Results apply within the tested circuit model, backend, qubits, calibration, shot counts, software harnesses, and stated experimental conditions. This is an experimental evidence record, not a universal security proof or production certification.”

This disclaimer is reproduced verbatim in the README and Zenodo deposit description.

---

9. MANIFEST Lock Reference

`MANIFEST.sha256` covers this preregistration document and all files under `src/`.
It is committed before any QPU job is submitted. Post-lock code changes are prohibited; any pre-execution harness defect triggers gate-stop protocol.

Remnant Fieldworks RF Process. Preregistration-first. No post-hoc changes.